

SONA COLLEGE OF TECHNOLOGY (Autonomous)

Department of Master of Computer Applications

**TOMORA AI: An explainable intelligent crop advisory platform
for tomato leaf disease diagnosis using deep learning**

Review - II

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PROJECT GUIDE	DESIGNATION
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OUTLINE:

- Problem Statement
- Abstract
- Objectives
- Existing System & Limitations
- Proposed System & Advantages
- Literature Study
- Hardware & Software Specifications
- System Design
- Dataset / Database Design
- Project Modules
- Screenshots
- References

PROBLEM STATEMENT

Tomato is one of the most widely cultivated vegetable crops, but it is highly susceptible to fungal, bacterial, and viral diseases. Manual disease identification often depends on farmers' experience and may lead to incorrect diagnosis and delayed treatment. Existing disease detection systems mainly focus on classification and provide limited support for crop management. Most of these systems lack Explainable AI, weather-based disease analysis, and multilingual assistance. Therefore, an intelligent crop advisory platform is required to provide accurate disease diagnosis and effective decision support for farmers.

ABSTRACT

Agriculture plays an important role in ensuring food security, where early disease detection is essential for improving crop productivity. This project proposes **TOMORA AI**, an Explainable AI-based crop advisory platform for tomato disease diagnosis using deep learning. The system employs a CNN model to identify diseases from tomato leaf images and uses Grad-CAM to visualize infected regions, improving transparency. It also integrates weather analysis, fertilizer recommendations, AI chatbot assistance, and bilingual support (English and Tamil). The proposed system aims to assist farmers with accurate disease detection and intelligent crop advisory through a user-friendly web application.

OBJECTIVES

- To develop an AI-based tomato leaf disease detection system using deep learning.
- To implement Explainable AI for visualizing disease prediction.
- To provide weather-based disease risk analysis and suitable fertilizer recommendations.
- To develop a bilingual web application (English & Tamil) for farmers.
- To design an intelligent AI chatbot for agricultural assistance and support.

EXISTING SYSTEM

Existing tomato leaf disease detection systems use deep learning models such as CNN to identify diseases from leaf images. However, they mainly focus on disease classification and provide limited support for complete crop management. **Most existing systems do not provide Explainable AI features to show the regions responsible for the prediction. They also lack weather-based disease risk analysis, multilingual assistance, and integrated agricultural recommendations.** Therefore, farmers may still need separate tools for diagnosis, treatment, and crop management.

LIMITATIONS OF EXISTING SYSTEM

- Low prediction reliability under real-world conditions.
- No Explainable AI support.
- Limited disease management recommendations.
- No weather-based disease prediction.
- Lack of multilingual interface.
- No integrated AI chatbot.
- Limited report generation and history management.

PROPOSED SYSTEM

TOMORA AI is a web-based crop advisory platform that detects tomato leaf diseases using CNN-based deep learning. It uses Explainable AI (Grad-CAM), weather-based disease prediction, and provides fertilizer, pesticide, and treatment recommendations with bilingual support, an AI chatbot, scan history, and PDF report generation.

Key Features

- Tomato leaf disease detection using Deep Learning (CNN).
- Explainable AI (Grad-CAM) for prediction visualization.
- Weather-based disease risk analysis.
- Fertilizer and pesticide recommendations.
- Bilingual interface (English & Tamil)

ADVANTAGES OF PROPOSED SYSTEM

- High disease detection accuracy using deep learning.
- Explainable AI improves transparency and user confidence.
- Weather-aware disease risk prediction.
- Intelligent fertilizer and pesticide recommendations.
- Bilingual interface (English & Tamil).
- PDF report generation.
- Easy to use for farmers.

LITERATURE STUDY

Name of the Authors	Paper Title	Journal details	Methodology	Observation
Fuentes, A., et al. (2023)	Deep Learning-Based Plant Disease Detection: A Review	Computers and Electronics in Agriculture, 2023	Reviewed CNN and Vision Transformer models for plant disease detection.	Deep learning provides accurate and reliable disease classification for crops.
Singh, P., et al. (2024)	Explainable AI for Plant Disease Detection Using Grad-CAM	IEEE Access, 2024	Applied Grad-CAM to visualize disease regions in leaf images.	Explainable AI improves model transparency and farmer trust.
Zhang, Y., et al. (2024)	Weather-Aware Deep Learning Framework for Crop Disease Prediction	Expert Systems with Applications, 2024	Combined weather data with deep learning for disease risk prediction.	Weather information significantly improves early disease detection accuracy.

Name of the Authors	Paper Title	Journal Details	Methodology	Observation
M. J. Jelali (2024)	Deep Learning Networks-Based Tomato Disease and Pest Detection: A First Review of Research Studies Using Real Field Datasets	<i>Frontiers in Plant Science</i> , 2024	Reviewed deep-learning-based tomato disease and pest detection studies using field and benchmark datasets, with emphasis on real agricultural conditions.	Many existing studies rely on controlled PlantVillage images; real-field datasets are important for improving robustness and practical deployment. (Frontiers)
R. H. Prince, A. A. Mamun, H. I. Peyal, S. Miraz, et al. (2024)	CSXAI: A Lightweight 2D CNN-SVM Model for Detection and Classification of Various Crop Diseases with Explainable AI Visualization	<i>Frontiers in Plant Science</i> , Vol. 15, 2024	Combined a lightweight 2D CNN-SVM model with Grad-CAM for crop disease classification and visual explanation.	Explainable AI visualization helps show the image regions responsible for predictions, improving transparency of disease classification models. (Frontiers)

HARDWARE SPECIFICATION

- **Processor:** 11th Gen Intel® Core
- **RAM:** 8 GB DDR4
- **Storage:** 256 GB SSD
- **Graphics:** Intel® UHD Graphics
- **System Type:** 64-bit Operating System, x64-based Processor
- **Camera:** Mobile Camera / Webcam
- **Internet Connection:** Broadband / Wi-Fi

SOFTWARE SPECIFICATION

- **Operating System:** Windows 11 (64-bit)
- **Programming Language:** Python 3.12+, JavaScript
- **Frontend:** HTML5, CSS3, JavaScript, Bootstrap 5
- **Backend Framework:** Flask
- **Deep Learning Framework:** TensorFlow, Keras (CNN)
- **Explainable AI:** Grad-CAM
- **Image Processing:** OpenCV, NumPy
- **Data Handling:** Pandas
- **Data Visualization:** Matplotlib
- **Cloud Storage:** Firebase Storage
- **Weather API:** OpenWeather API



TOMORA AI

Project Timeline

From Leaf to Life • Smarter Farming • Healthier Tomorrow

*Small Steps
Big Impact*



S.No	Task	JUL 2026	AUG 2026	SEP 2026	OCT 2026	NOV 2026
1	 Literature Survey					
2	 Requirement Analysis					
3	 Data Collection					
4	 System Design					
5	 Development					
6	 AI Model Integration					
7	 Testing					
8	 Documentation & Final Submission					

SYSTEM DESIGN

TOMORA AI — SYSTEM ARCHITECTURE

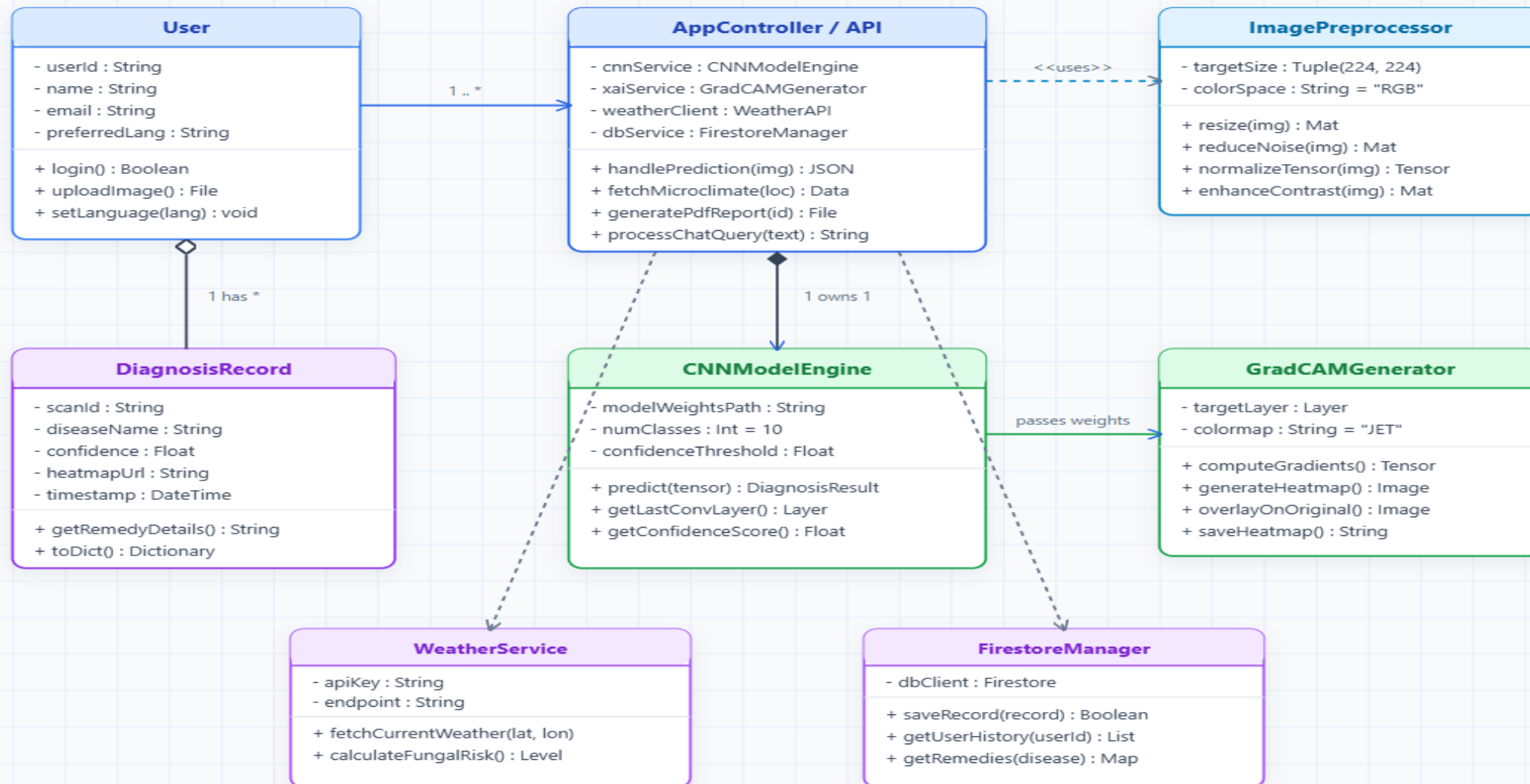
AI-Based Tomato Disease Detection • Explainable AI • Weather Intelligence • Smart Crop Advisory



SYSTEM DESIGN

UML CLASS DIAGRAM

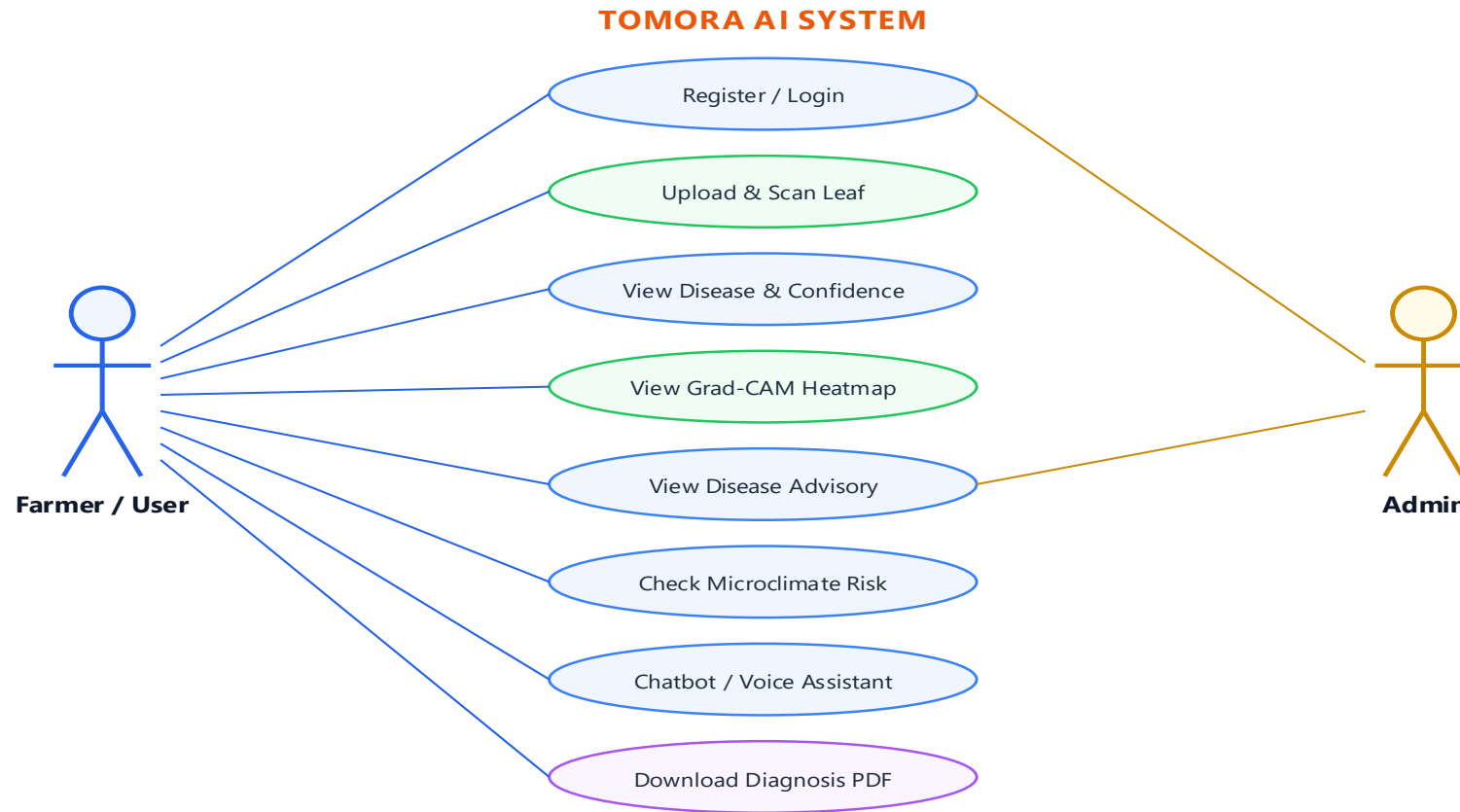
TOMORA AI — Software Architecture & Object Relationships



SYSTEM DESIGN

USE CASE DIAGRAM

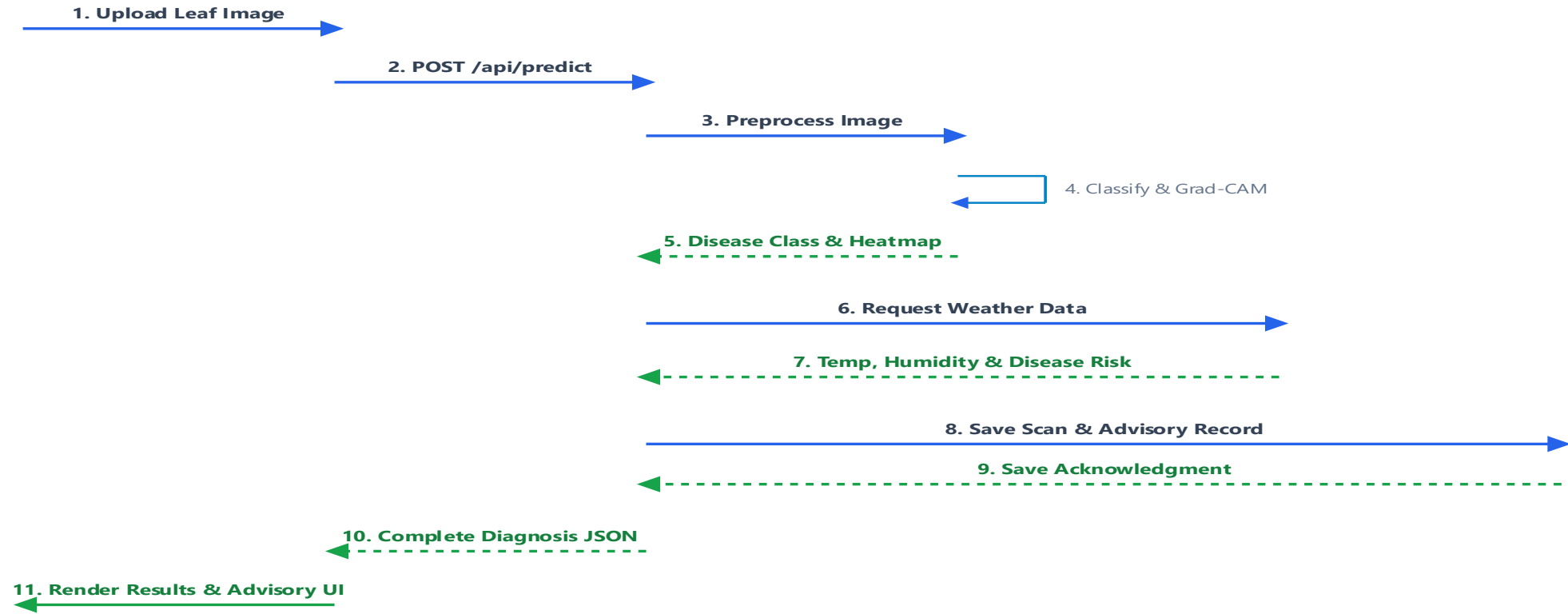
TOMORA AI Platform Interactions



SYSTEM DESIGN

SYSTEM SEQUENCE DIAGRAM

TOMORA AI - Leaf Upload to Diagnosis & Weather Assessment Flow



DATASET OR DATABASE DESIGN

USERS

- User ID
- Email / Login
- Role
- Language

PREDICTION HISTORY

- Scan ID
- User ID
- Image URL
- Disease
- Confidence
- Timestamp

DISEASE ADVISORY

- Disease
- Symptoms
- Treatment
- Prevention
- Fertilizer
- Pesticide

WEATHER DATA

- Record ID
- Location
- Temperature
- Humidity
- Rainfall
- Risk

CHAT HISTORY

- Chat ID
- User ID
- Question
- Response
- Timestamp

REPORTS

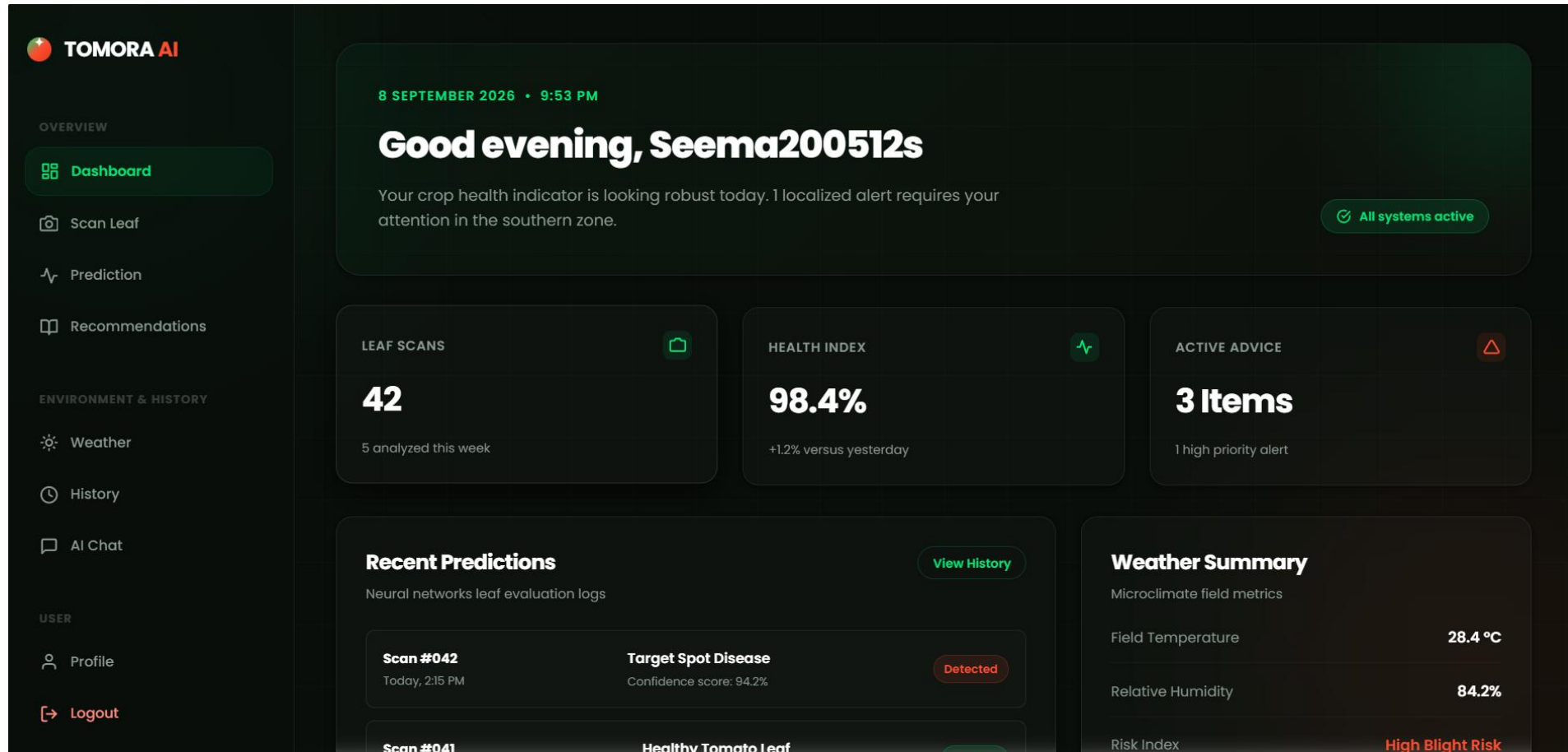
- Report ID
- User ID
- Scan ID
- Report URL
- Created At

User → Prediction History → Advisory / Weather → Chat → Report

PROJECT MODULES

- 1. User Authentication – registration, login/logout, protected dashboard.
- 2. Leaf Disease Detection – upload, preprocessing, CNN classification, confidence.
- 3. Explainable AI – Grad-CAM heatmap and prediction explanation.
- 4. Disease Advisory – symptoms, treatment, prevention, fertilizer and pesticide.
- 5. Weather Intelligence – weather parameters and disease-risk assessment.
- 6. AI Chat Assistant – crop/disease assistance, English/Tamil and voice support.
- 7. History & Reporting – previous scans, delete/view and PDF diagnosis report.

SCREEN SHOTS



SCREEN SHOTS

 **TOMORA AI**

OVERVIEW

Dashboard

Scan Leaf

Prediction

Recommendations

ENVIRONMENT & HISTORY

Weather

History

AI Chat

USER

Profile

Logout

Leaf Scanner Portal

Upload leaf snapshots to trigger explainable computer vision diagnostics.



Camera Capture

Run AI Scan

Diagnostic Guidelines

For high-accuracy results, please ensure the photo adheres to standard guidelines:

1

Single Leaf Center

Focus on a single leaf containing the anomaly in the center of the frame.

2

Adequate Lighting

Capture the image under bright daylight or normal ambient light. Avoid shadows.

3

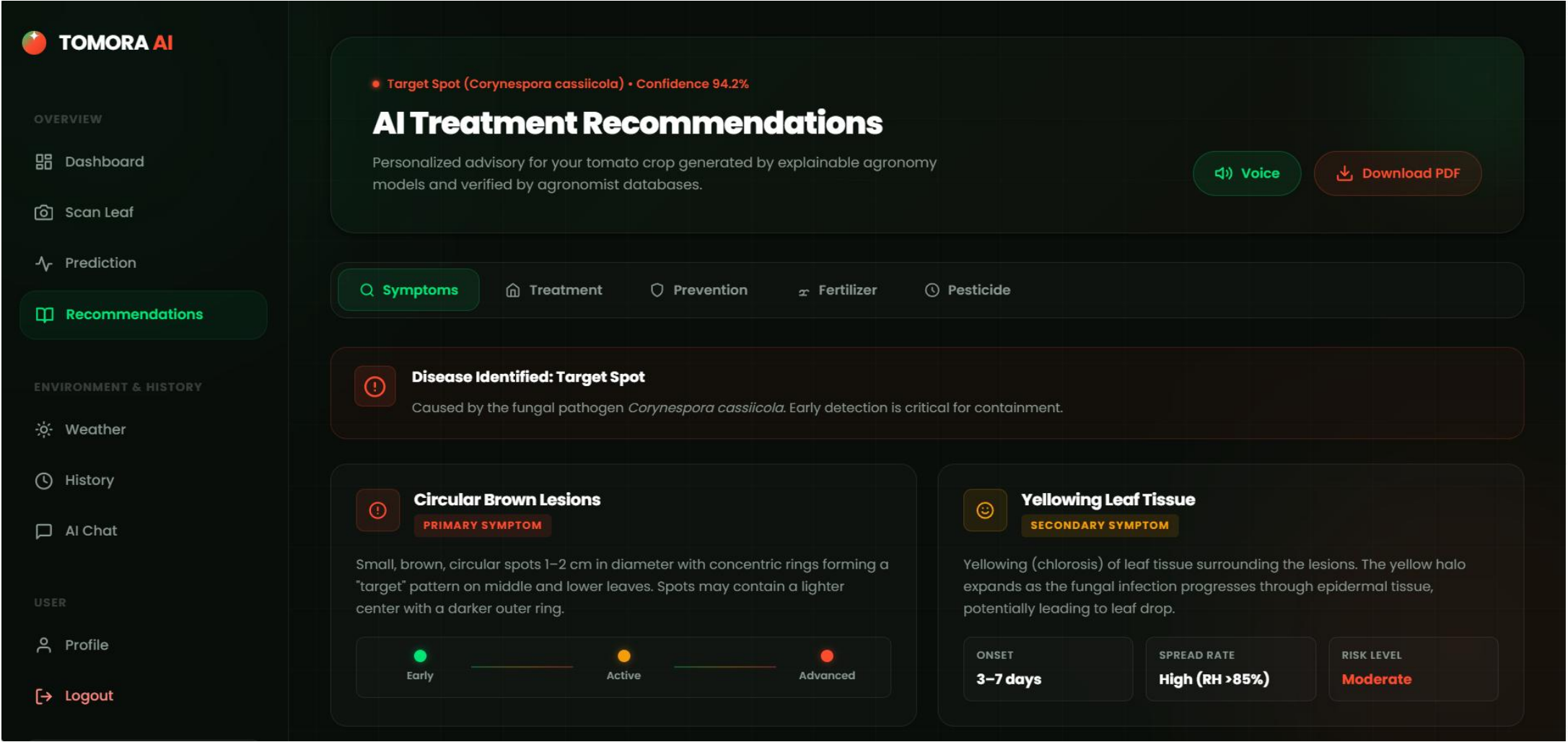
Clear Focus

Ensure the spots, margins, or blights are in sharp focus (no motion blur).

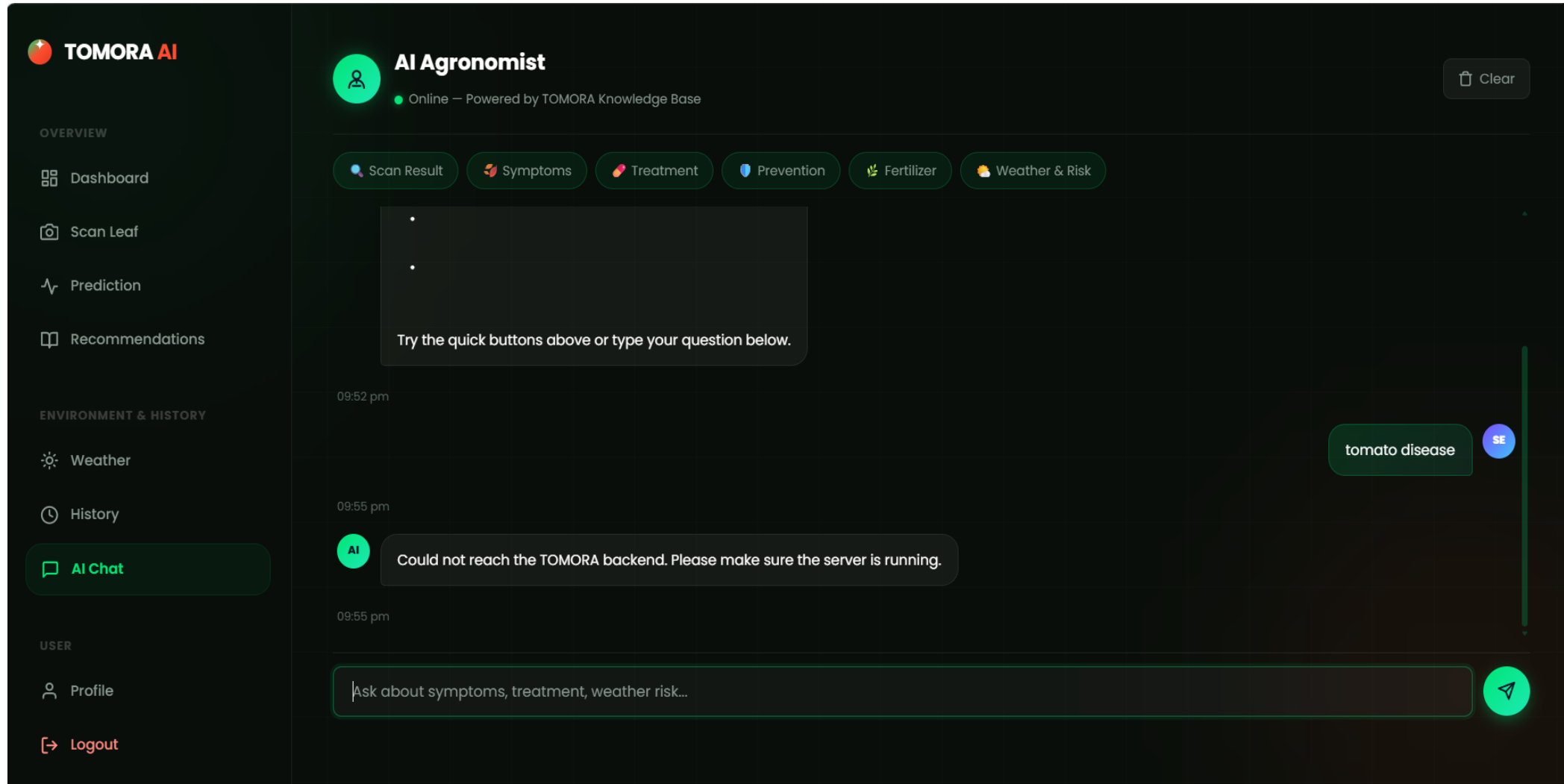
09-09-2026

22

SCREEN SHOTS



SCREEN SHOTS



REFERENCES

1. M. Abdul et al., "Deep Learning-Based Classification, Detection, and Segmentation of Tomato Diseases: A Comprehensive Review," Smart Agricultural Technology, vol. 10, 2025.
2. J. Wang et al., "Deep Learning Networks-Based Tomato Disease and Pest Detection: A Comprehensive Review," Frontiers in Plant Science, vol. 15, 2024.
3. X. Liu et al., "An Efficient Deep Learning Model for Tomato Disease Detection," Plant Methods, vol. 20, 2024.
4. M. Hassan et al., "Exploring Explainable AI Techniques for Plant Disease Classification in Smart Agriculture," Procedia Computer Science, vol. 252, 2025.

THANK YOU